

e commerce

Req2Design · IEEE 830-1998 aligned

Document Control

Version	Date	Author	Change Summary
1.0	2026-04-09	ali	Initial generated draft for structured review.

Document ID: SRS-20260409-181702

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INTRODUCTION

Purpose: To define the requirements for an e-commerce platform that allows users to browse, purchase, and administrators to manage products and orders.

Scope: This document covers the functionalities required for browsing, purchasing, managing products, handling payments, and processing returns and refunds. It does not cover aspects related to security, scalability, maintainability, or availability.

Definitions/Acronyms: E-commerce

- Electronic Commerce; GUI
- Graphical User Interface; HTTP
- Hypertext Transfer Protocol; HTML
- HyperText Markup Language; CSS
- Cascading Style Sheets; SQL
- Structured Query Language; API
- Application Programming Interface.

References: IEEE Std 830-1998, IEEE Standard for Software Requirements Specifications.

Overview: The e-commerce application will provide a web-based interface for both users and administrators to interact with the product catalog, place orders, and manage transactions.

OVERALL DESCRIPTION

Product Perspective: The e-commerce platform is designed to cater to online shopping needs, offering a seamless experience through a responsive and accessible website. The primary focus is on ease of use and secure transaction management.

Product Functions:

- Users can browse products, search by keywords and filters, add items to cart, and checkout.
- Administrators can add, edit, and remove products, manage inventory, and review orders.
- Secure payment processing with receipt emails.
- Support for promotional discounts and coupon codes.
- Graceful handling of returns, refunds, and out-of-stock items.

User Characteristics: Target audience includes general consumers looking to shop online and business owners/administrators needing tools to manage their e-commerce operations.

Constraints:

- Must be compatible across different devices and browsers.
- Responsive design to ensure mobile compatibility.
- Secure payment gateway integration.

Assumptions/Dependencies:

- Internet connectivity for all interactions.

- Availability of backend servers and databases.

- Validity of entered product information by administrators.

SPECIFIC REQUIREMENTS

External Interface Requirements:

- **User Interface:** A clean, intuitive, and responsive GUI for easy navigation and interaction.
- **Hardware Interface:** Compatible with various devices including desktops, laptops, tablets, and smartphones.
- **Software Interface:** Supports modern web browsers and operating systems.
- **Communication Interface:** Utilizes standard protocols such as HTTP and HTTPS for secure data transmission.

Functional Requirements:

- **FR-01:** Browse Products

Input: None

Processing: Display list of available products sorted by category or relevance.

Output: List of products with images, descriptions, prices, and ratings.

Priority: High

- **FR-02:** Search Products

Input: Keywords or filter criteria.

Processing: Filter and display relevant products based on provided search terms or selected filters.

Output: Filtered list of products matching the query.

Priority: Medium

- **FR-03:** Add Items to Cart

Input: Selected product ID and quantity.

Processing: Update local storage or session state with added items.

Output: Confirmation message and updated cart summary.

Priority: High

- **FR-04:** Checkout Process

Input: Cart contents and shipping details.

Processing: Validate inputs, calculate total cost, process payment, generate order confirmation.

Output: Order summary and receipt via email.

Priority: High

- **FR-05:** Manage Products (for Administrators)

Input: New product details, existing product IDs, and update criteria.

Processing: Insert new entries, modify existing records, delete specified entries from database.

Output: Updated product listings and success messages.

Priority: High

- **FR-06:** Inventory Management

Input: Stock levels and other inventory-related data.

Processing: Track stock changes and update database accordingly.

Output: Real-time updates on product availability.

Priority: Medium

- **FR-07:** Review Orders

Input: Order ID(s) or customer name(s).

Processing: Retrieve detailed order history and associated customer information.

Output: Comprehensive order reports.

Priority: Low

Non-

functional Requirements:

- **Usability:** Intuitive and user-friendly interface.

- **Reliability:** Minimal downtime and consistent performance.

- **Performance:** Fast response times and efficient processing of transactions.

- **Portability:** Cross-platform compatibility ensuring accessibility on multiple devices.

SYSTEM FEATURES

Feature 1: Product Browsing

- Provide categorized views of products along with filtering options.
- Implement pagination for large datasets.
- Enable sorting by price, rating, popularity, etc.

Feature 2: Search Functionality

- Allow keyword searches within product titles, descriptions, and attributes.
- Offer advanced filtering based on price range, brand, color, size, etc.

Feature 3: Shopping Cart Integration

- Save cart contents locally until checkout or expiration.
- Display real-time totals and allow adjustments before finalizing purchase.

Feature 4: Payment Gateway

- Integrate with trusted third-party services for secure credit card transactions.
- Send automated receipts upon successful payment completion.

Feature 5: Administrator Tools

- Admin panel for adding, editing, deleting product listings.
- Inventory tracking dashboard showing current stock levels.
- Order management section allowing viewing and updating of past sales.

Feature 6: Promotions & Discounts

- Apply percentage off or fixed amount reductions during checkout.
- Use promo codes to offer special deals to customers.

Feature 7: Return Policy Handling

- Automatically detect when items go out of stock and notify users.
- Maintain a log of return requests and refund status.
- Ensure that returned goods do not affect ongoing cart sessions.

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